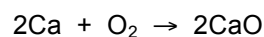




REACTING MASS CALCULATIONS 3

- 1) Calculate the mass of calcium that can react with 40 g of oxygen.

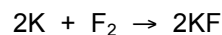


$$\text{moles O}_2 = \frac{\text{mass}}{M_r} = \frac{40}{32} = 1.25 \text{ mol}$$

$$\text{moles Ca} = 2 \times 1.25 = 2.50 \text{ mol}$$

$$\text{mass Ca} = M_r \times \text{moles} = 40 \times 2.50 = 100 \text{ g}$$

- 2) Calculate the mass of fluorine that reacts with 3.9 g of potassium.

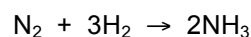


$$\text{moles K} = \frac{\text{mass}}{M_r} = \frac{3.9}{39} = 0.10 \text{ mol}$$

$$\text{moles F}_2 = \frac{0.10}{2} = 0.050 \text{ mol}$$

$$\text{mass F}_2 = M_r \times \text{moles} = 38 \times 0.050 = 1.9 \text{ g}$$

- 3) Calculate the mass of nitrogen that reacts with 30 g of hydrogen.

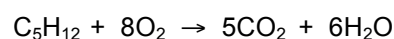


$$\text{moles H}_2 = \frac{\text{mass}}{M_r} = \frac{30}{2} = 15 \text{ mol}$$

$$\text{moles N}_2 = \frac{15}{3} = 5 \text{ mol}$$

$$\text{mass N}_2 = M_r \times \text{moles} = 28 \times 5 = 140 \text{ g}$$

- 4) What mass of carbon dioxide is made when 7.2 g of pentane (C₅H₁₂) burns in oxygen?

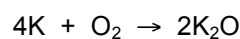


$$\text{moles C}_5\text{H}_{12} = \frac{\text{mass}}{M_r} = \frac{7.2}{72} = 0.10 \text{ mol}$$

$$\text{moles CO}_2 = 5 \times 0.10 = 0.50 \text{ mol}$$

$$\text{mass CO}_2 = M_r \times \text{moles} = 44 \times 0.5 = 22 \text{ g}$$

- 5) What mass of potassium can react with 4.0 g of oxygen?

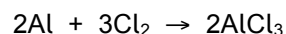


$$\text{moles O}_2 = \frac{\text{mass}}{M_r} = \frac{4.0}{32} = 0.125 \text{ mol}$$

$$\text{moles K} = 4 \times 0.125 = 0.50 \text{ mol}$$

$$\text{mass K} = M_r \times \text{moles} = 39 \times 0.5 = 19.5 \text{ g}$$

6) What mass of chlorine reacts with 8.1 g of aluminium?

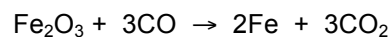


$$\text{moles Al} = \frac{\text{mass}}{M_r} = \frac{8.1}{27} = 0.30 \text{ mol}$$

$$\text{moles Cl}_2 = \frac{3}{2} \times 0.30 = 0.45 \text{ mol}$$

$$\text{mass Cl}_2 = M_r \times \text{moles} = 71 \times 0.45 = 31.95 \text{ g}$$

7) What mass of iron can be made from 20 kg of iron(III) oxide?

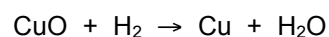


$$\text{moles Fe}_2\text{O}_3 = \frac{\text{mass}}{M_r} = \frac{20000}{160} = 125 \text{ mol}$$

$$\text{moles Fe} = 2 \times 125 = 250 \text{ mol}$$

$$\text{mass Fe} = M_r \times \text{moles} = 56 \times 250 = 14000 \text{ g}$$

8) What mass of hydrogen is needed to react with 31.8 mg of copper(II) oxide?

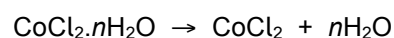


$$\text{moles CuO} = \frac{\text{mass}}{M_r} = \frac{0.0318}{79.5} = 0.00040 \text{ mol}$$

$$\text{moles H}_2 = 0.00040 \text{ mol}$$

$$\text{mass H}_2 = M_r \times \text{moles} = 2 \times 0.00040 = 0.00080 \text{ g}$$

9) 5.95 g of hydrated cobalt(II) chloride decompose to form 3.25 g of anhydrous cobalt(II) chloride on heating. Calculate the formula mass of hydrated cobalt(II) chloride and the value of n .



$$\text{moles CoCl}_2 = \frac{\text{mass}}{M_r} = \frac{3.25}{130} = 0.025 \text{ mol}$$

$$\text{mass H}_2\text{O} = 5.95 - 3.25 = 2.70 \text{ g}$$

$$\text{moles H}_2\text{O} = \frac{\text{mass}}{M_r} = \frac{2.70}{18} = 0.15 \text{ mol}$$

$$\text{Ratio of moles CoCl}_2 : \text{H}_2\text{O} = 0.025 : 0.15 = \frac{0.025}{0.025} : \frac{0.150}{0.025} = 1 : 6$$

$$\therefore n = 6 \text{ (nearest whole number)}$$

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Can convert units		
Shows suitable working			Can use reacting ratios in equations			Can find water of crystallisation		
Can work out M_r			Can find mass from moles			Gives units		